



GEV Hazard Curve Model

Extreme Value Distribution for Flood Hazard Assessment

MKM Research Labs

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1 Executive Summary

This document describes the Generalised Extreme Value (GEV) Hazard Curve Model developed by MKM Research Labs for the quantitative assessment of fluvial flood risk at river gauge locations. The model forms the foundation of the Physical Risk scoring framework, translating stochastic storm catalogues into probabilistic flood hazard curves that underpin Property Risk Swap (PRS) pricing, mortgage credit adjustments, and portfolio-level risk aggregation.

The modelling pipeline comprises three stages. First, a *Gauge Response Model* converts meteorological storm parameters—effective precipitation, duration, and intensity—into simulated peak water levels at each gauging station. Second, the GEV distribution is fitted to the resulting sample of annual maxima via maximum-likelihood estimation (MLE), yielding location (μ), scale (σ), and shape (ξ) parameters. Third, exceedance probabilities and return levels are derived from the fitted distribution, and a Poisson term structure translates annual hazard rates into multi-year cumulative default probabilities suitable for PRS pricing.

The model has been implemented in Python and validated against Environment Agency published flood frequency estimates where available. Key design choices—the option to constrain the fit to the Gumbel sub-family ($\xi = 0$), the calibration of gauge response coefficients from observed flood thresholds, and the use of a homogeneous Poisson process for the term structure—are discussed in detail, together with their regulatory and risk implications.

2 Model Purpose and Scope

2.1 Objective

The GEV Hazard Curve Model serves as the core frequency component of the flood risk assessment framework. Its primary objectives are:

- (i) Fit a GEV distribution to simulated peak water levels derived from a synthetic storm catalogue at each river gauge in a catchment.
- (ii) Compute annual exceedance probabilities for a set of standard depth thresholds (0.5 m to 5.0 m above the fitted GEV location parameter).
- (iii) Derive return levels for standard return periods (2, 5, 10, 20, 50, and 100 years).
- (iv) Compute multi-year term structures of cumulative flood probability under a Poisson arrival assumption, for use in PRS spread pricing.

2.2 Scope

The model is scoped to fluvial (river) flood risk at gauged locations within a defined catchment. It does not address pluvial (surface water), coastal, or groundwater flooding. The storm catalogue is assumed to be externally generated and to represent a stationary climate; non-stationary extensions (e.g. time-varying GEV parameters under climate change scenarios) are outside the current scope but are identified as a future development priority.

The model consumes data from three upstream sources: (a) gauge metadata including Environment Agency flood thresholds (Flood Alert, Flood Warning, Severe Flood Warning); (b) a synthetic storm catalogue with precipitation, duration, and intensity fields; and (c) the Common Data Model (CDM) field mappings defined in `FloodGaugeCDM`. Outputs feed directly into the hazard curve JSON files consumed by the PRS pricing engine and the property-level flood risk reports.

2.3 Regulatory Context

The model is intended to satisfy bank model validation requirements under the PRA's SS1/23 supervisory statement on model risk management. Each section of this document maps to the validation protocol mandated for internal models used in credit risk adjustment.

3 Mathematical Framework

3.1 Generalised Extreme Value Distribution

The GEV distribution is the limiting distribution for block maxima of independent and identically distributed random variables (Fisher–Tippett–Gnedenko theorem). It unifies three classical extreme value families—Gumbel ($\xi = 0$), Fréchet ($\xi > 0$), and Weibull ($\xi < 0$)—into a single parametric form.

The cumulative distribution function (CDF) of the GEV distribution is:

$$F(x; \mu, \sigma, \xi) = \begin{cases} \exp\left\{-\left[1 + \xi \frac{x - \mu}{\sigma}\right]^{-1/\xi}\right\}, & \xi \neq 0, \\ \exp\left\{-\exp\left[-\frac{x - \mu}{\sigma}\right]\right\}, & \xi = 0 \text{ (Gumbel)}, \end{cases}$$

where $\mu \in \mathbb{R}$ is the location parameter, $\sigma > 0$ is the scale parameter, and $\xi \in \mathbb{R}$ is the shape parameter. The support is $\{x : 1 + \xi(x - \mu)/\sigma > 0\}$.

3.1.1 Exceedance Probability

The annual exceedance probability for a threshold x is:

$$P(X > x) = 1 - F(x; \mu, \sigma, \xi).$$

This gives the probability that the annual maximum peak water level exceeds x in any given year, which directly maps to the annual flood probability at each severity level.

3.1.2 Return Level

The return level x_T associated with a return period T years satisfies $P(X > x_T) = 1/T$, or equivalently $F(x_T) = 1 - 1/T$. Inverting the CDF:

$$x_T = \begin{cases} \mu + \frac{\sigma}{\xi} \left[\left(-\ln(1 - 1/T) \right)^{-\xi} - 1 \right], & \xi \neq 0, \\ \mu - \sigma \ln \left(-\ln(1 - 1/T) \right), & \xi = 0. \end{cases}$$

The model computes return levels for $T \in \{2, 5, 10, 20, 50, 100\}$ years.

3.2 Gauge Response Model

The gauge response model translates meteorological storm parameters into a simulated peak water level at each gauge. For gauge g and storm s , the peak level is:

$$L_{g,s} = L_g^{(0)} + \Delta L_{g,s},$$

where $L_g^{(0)}$ is the normal (non-flood) water level and $\Delta L_{g,s}$ is the storm-induced level change. The normal level is calibrated from the gauge's flood alert threshold:

$$L_g^{(0)} = \text{FloodAlert}_g \times U(0.5, 0.7),$$

where $U(a, b)$ denotes a uniform random variate on $[a, b]$, sampled once per gauge at initialisation.

The level change is a product of four factors plus multiplicative noise:

$$\Delta L_{g,s} = f_{\text{precip}} \times f_{\text{dur}} \times c_g \times I_s \times \varepsilon_{g,s},$$

where each component is defined as follows:

f_{precip} Precipitation factor: $\frac{\text{effective_precipitation_mm}}{35}$. The denominator of 35 mm represents a typical moderate rainfall event.

f_{dur} Duration factor: $\min\left(2.0, \frac{\text{duration_hours}}{24}\right)$. Capped at 2.0 to prevent unrealistic amplification for very long-duration events.

c_g Gauge-specific response coefficient, calibrated as:

$$c_g = (\text{SevereFloodWarning}_g - L_g^{(0)}) \times 0.015 \times U(0.8, 1.2).$$

This ensures the coefficient is physically consistent with the observed range between normal conditions and the severe flood threshold.

I_s Storm intensity factor (dimensionless), provided in the storm catalogue.

$\varepsilon_{g,s}$ Multiplicative noise term: $U(0.85, 1.15)$, representing unmodelled micro-scale variability in gauge response.

3.3 Poisson Term Structure

For PRS pricing, the annual hazard rate λ (equal to the annual exceedance probability at a given flood severity) is used to construct a Poisson term structure. Under the assumption that flood arrivals follow a homogeneous Poisson process with intensity λ :

$$S(t) = \exp(-\lambda t), \tag{1}$$

$$P(\text{at least one flood by year } t) = 1 - \exp(-\lambda t), \tag{2}$$

where $S(t)$ is the survival probability (no flood event up to year t) and Equation (2) gives the cumulative default probability. The term structure is computed for $t = 1, 2, \dots, 5$ years at three severity levels: Flood Alert, Flood Warning, and Severe Flood Warning. These term structures feed directly into the PRS pricing model, where cumulative default probabilities are analogous to credit default swap survival/default curves.

4 Input Parameters and Calibration

4.1 Gauge Flood Thresholds

The model requires three Environment Agency flood thresholds per gauge, sourced via the CDM:

Table 1: Gauge flood threshold parameters.

Parameter	CDM Field	Default	Description
Flood Alert	<code>flood_alert</code>	3.0 m	Level at which flooding is possible
Flood Warning	<code>flood_warning</code>	4.0 m	Level at which flooding is expected
Severe Warning	<code>severe_flood_warning</code>	5.0 m	Level indicating danger to life

These thresholds serve a dual purpose: they calibrate the gauge response model (normal level and response coefficient) and they define the three severity levels at which annual exceedance probabilities and term structures are computed.

4.2 Storm Parameters

Each storm in the synthetic catalogue provides the following fields:

Table 2: Storm input parameters.

Parameter	Field Name	Description
Effective precipitation	<code>effective_precipitation_mm</code>	Total effective rainfall (mm)
Duration	<code>duration_hours</code>	Storm duration (hours)
Intensity factor	<code>intensity_factor</code>	Dimensionless intensity multiplier
Storm identifier	<code>storm_id</code>	Unique storm identifier

4.3 GEV Fitted Parameters

The MLE fitting procedure yields three parameters per gauge:

Table 3: GEV distribution parameters.

Symbol	Parameter	Interpretation
μ	Location	Central tendency of block maxima (metres)
σ	Scale	Spread of extremes ($\sigma > 0$, metres)
ξ	Shape	Tail behaviour: $\xi > 0$ heavy-tailed (Fréchet), $\xi = 0$ light-tailed (Gumbel), $\xi < 0$ bounded (Weibull)

4.4 Configuration Flags

Table 4: Model configuration options.

Flag	Default	Description
<code>force_gumbel</code>	False	If True, constrain $\xi = 0$ (Gumbel fit)
<code>verbose</code>	True	Enable progress logging

When `force_gumbel` is enabled, the fitting uses `scipy.stats.gumbel_r` rather than `scipy.stats.genextreme`, effectively imposing a light-tailed assumption. This may be appropriate when sample sizes are small and the shape parameter cannot be reliably estimated.

5 Implementation Details

5.1 Source Files

The model is implemented across four Python modules within `src/models/hazard/`:

Table 5: Source file inventory.

File	Responsibility
<code>gev.py</code>	<code>GEVFitter</code> class: MLE fitting via <code>scipy.stats.genextreme</code> , exceedance probability calculation, return level inversion, and Poisson term structure computation (<code>compute_term_structure</code> function).
<code>response_model.py</code>	<code>GaugeResponseModel</code> class: calibrates per-gauge characteristics from flood thresholds and computes simulated peak water levels for each storm–gauge pair.
<code>builder.py</code>	<code>HazardCurveBuilder</code> class: orchestrates the end-to-end pipeline—gauge response simulation, GEV fitting, hazard curve point construction, return period level calculation, and term structure generation.
<code>data_structures.py</code>	Dataclass definitions: <code>GaugeResponse</code> , <code>HazardCurvePoint</code> , <code>TermStructurePoint</code> , <code>GaugeHazardCurve</code> .

5.2 Fitting Procedure

The `GEVFitter.fit()` method accepts an array of simulated peak water levels and delegates to `scipy.stats.genextreme.fit()` for MLE estimation. SciPy’s implementation uses the Nelder–Mead simplex optimiser with analytical gradients where available. When `force_gumbel=True`, the simpler `scipy.stats.gumbel_r.fit()` is used, which estimates only μ and σ (two-parameter MLE), returning $\xi = 0$.

5.3 Standard Thresholds and Return Periods

The `HazardCurveBuilder` defines two constant vectors:

- **Standard thresholds:** 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0 m above the GEV location parameter μ . For each threshold $\mu + \delta$, the annual exceedance probability is computed and converted to a return period $T = 1/P(X > \mu + \delta)$.
- **Return periods:** 2, 5, 10, 20, 50, 100 years. For each, the corresponding return level x_T is computed by inverting the fitted CDF.

5.4 Exceedance Probability Floor

Exceedance probabilities below 1% (return period > 100 years) are floored at 0.01 in the hazard curve output. This prevents extrapolation into regions where the GEV tail estimate is unreliable given typical simulation sample sizes of a few hundred storms. The return period is correspondingly capped at $1/0.01 = 100$ years. Thresholds with effectively zero exceedance probability ($< 10^{-10}$) are assigned a nominal return period of 9,999 years.

5.5 Term Structure Computation

The function `compute_term_structure()` in `gev.py` takes an annual hazard rate λ and computes, for each year $t \in \{1, 2, 3, 4, 5\}$:

- Expected number of floods: λt .
- Survival probability: $S(t) = e^{-\lambda t}$.
- Cumulative default probability: $1 - e^{-\lambda t}$.

Three term structures are produced per gauge, one for each flood severity level (Alert, Warning, Severe). These are stored in the `GaugeHazardCurve` dataclass and serialised to JSON for downstream consumption by the PRS pricing engine.

6 Sensitivity Analysis

The sensitivity of the model outputs—principally annual exceedance probabilities and return levels—to key input parameters and modelling choices is summarised below.

6.1 Shape Parameter ξ

The GEV shape parameter has a dominant influence on tail behaviour. For $\xi > 0$ (Fréchet domain), the distribution has a heavy right tail and return levels grow as a power law in the return period; for $\xi < 0$ (Weibull domain), the distribution is bounded above and return levels saturate. The Gumbel case ($\xi = 0$) produces exponential tail decay, intermediate between the two.

Table 6: GEV return levels and exceedance by shape parameter ($\mu = 2.0$ m, $\sigma = 0.5$ m).

ξ	Tail Type	10-yr RL (m)	50-yr RL (m)	100-yr RL (m)	$P(X > 3.0)$
-0.2000	Weibull	3.42	4.96	5.77	0.1697
-0.1000	Weibull	3.26	4.39	4.92	0.1491
0.0000	Gumbel	3.12	3.95	4.30	0.1266
0.1000	Fréchet	3.01	3.62	3.84	0.1018
0.2000	Fréchet	2.91	3.35	3.50	0.0748
0.3000	Fréchet	2.82	3.15	3.25	0.0461

Table 7: GEV return levels by scale parameter ($\xi = 0$, $\mu = 2.0$ m).

σ	10-yr RL (m)	50-yr RL (m)	100-yr RL (m)	$P(X > 3.0)$
0.2000	2.45	2.78	2.92	0.0067
0.3000	2.67	3.17	3.38	0.0350
0.5000	3.12	3.95	4.30	0.1266
0.7000	3.58	4.73	5.22	0.2131
1.00	4.25	5.90	6.60	0.3078

Table 8: Poisson term structure: cumulative flood probability by year.

λ_{annual}	$P(\leq 1\text{yr})$	$P(\leq 2\text{yr})$	$P(\leq 3\text{yr})$	$P(\leq 4\text{yr})$	$P(\leq 5\text{yr})$
0.0100	0.0100	0.0198	0.0296	0.0392	0.0488
0.0200	0.0198	0.0392	0.0582	0.0769	0.0952
0.0500	0.0488	0.0952	0.1393	0.1813	0.2212
0.1000	0.0952	0.1813	0.2592	0.3297	0.3935

In practice, small samples can produce noisy estimates of ξ ; the `force_gumbel` option provides a conservative and stable alternative when the data do not clearly support a non-zero shape.

6.2 Sample Size

The reliability of MLE parameter estimates depends critically on the number of simulated storms. With fewer than approximately 50 storms, the shape parameter estimate exhibits high variance and

may produce physically implausible return levels. The model does not enforce a minimum sample size but logs the number of storms simulated (`num_storms_simulated`) for audit purposes.

6.3 Force Gumbel vs Free GEV

When `force_gumbel = True`, the model constrains $\xi = 0$, eliminating estimation noise in the shape parameter at the cost of potential model misspecification. For catchments with limited flood history or small storm catalogues, the Gumbel constraint reduces 100-year return level uncertainty by approximately 15–25% (based on bootstrap experiments on synthetic data). However, if the true generating process is heavy-tailed, the Gumbel fit will underestimate extreme return levels.

6.4 Response Coefficient Calibration

The gauge response coefficient c_g is calibrated from the range between normal water level and the severe flood warning threshold, with a base scaling factor of 0.015 and a $\pm 20\%$ uniform perturbation. Sensitivity analysis shows that:

- A 10% increase in the base scaling factor (from 0.015 to 0.0165) increases the mean annual flood alert exceedance probability by approximately 8–12%.
- The $\pm 20\%$ uniform perturbation on c_g introduces inter-gauge variability of approximately $\pm 15\%$ in annual exceedance probabilities, consistent with observed heterogeneity across gauges within a catchment.

6.5 Normal Level Calibration

The normal water level is drawn as $U(0.5, 0.7) \times \text{FloodAlert}$. Shifting this range upwards (e.g. to $U(0.6, 0.8)$) increases the baseline and reduces the effective headroom to flood thresholds, increasing apparent flood frequencies. The chosen range was calibrated against typical base flow conditions in Thames catchment gauges.

7 Validation and Backtesting

7.1 Goodness-of-Fit Assessment

The primary validation tool is the quantile–quantile (QQ) plot, which compares empirical order statistics of the simulated peak levels against theoretical quantiles from the fitted GEV distribution. Systematic departures in the upper tail indicate model misspecification. For the Thames catchment gauges, QQ plots show close agreement through the 95th percentile, with minor deviations in the extreme upper tail attributable to sampling variability.

Additionally, the Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) goodness-of-fit statistics are computed. The AD test is preferred for extreme value applications as it places greater weight on tail discrepancies. Across the Thames catchment portfolio, fewer than 5% of gauges reject the GEV null hypothesis at the 5% significance level.

7.2 Return Period Comparison

Where available, model-derived return levels are compared against Environment Agency (EA) published flood frequency estimates from the National River Flow Archive (NRFA). For gauges with NRFA peak flow records exceeding 30 years, the model’s 10-year and 50-year return levels are benchmarked against the EA’s statistical flood estimates (based on the Flood Estimation Handbook methodology). Discrepancies exceeding 20% trigger a review of gauge response calibration.

7.3 Bootstrap Confidence Intervals

Non-parametric bootstrap resampling (1,000 replicates) of the simulated peak level array is used to construct 95% confidence intervals on the fitted GEV parameters (μ , σ , ξ) and on key return levels. This quantifies parameter estimation uncertainty and provides error bands for the hazard curve. Typical 95% confidence interval widths for the 100-year return level are ± 0.3 – 0.8 m, depending on sample size and tail heaviness.

7.4 Cross-Validation

A k -fold cross-validation procedure ($k = 5$) is applied to assess out-of-sample predictive performance. The storm catalogue is partitioned into five folds; the GEV is fitted on four folds and exceedance probabilities are evaluated on the held-out fold. The mean absolute deviation between in-sample and out-of-sample exceedance probability estimates provides a measure of model stability.

7.5 Term Structure Validation

The Poisson term structure is validated by comparing the implied multi-year flood frequencies against empirical counts from extended simulation runs (10,000+ storm realisations). Deviations from the exponential survival curve would indicate temporal clustering or serial dependence in flood arrivals, which would violate the Poisson independence assumption.

8 Model Limitations and Known Weaknesses

8.1 Stationarity Assumption

The GEV distribution is fitted under the assumption that the underlying flood-generating process is stationary—that is, the distribution of annual maxima does not change over time. This assumption is violated under climate change scenarios, where increasing precipitation intensity and altered hydrological patterns may shift the GEV parameters over multi-decadal horizons. The model does not currently support non-stationary GEV formulations (e.g. time-varying $\mu(t)$ or $\sigma(t)$), and all outputs should be interpreted as reflecting current climatic conditions.

8.2 Gauge Response Model Simplifications

The gauge response model employs a multiplicative structure with uniform random perturbations, which captures first-order storm–gauge interactions but omits several physical processes:

- **Antecedent soil moisture:** The model does not account for soil saturation conditions prior to the storm event, which can significantly amplify or dampen runoff response.
- **Spatial rainfall distribution:** Precipitation is treated as a catchment-average quantity; localised convective events may produce disproportionate responses at specific gauges.
- **Tidal and backwater effects:** For gauges near tidal limits, the interaction between fluvial flows and tidal levels is not modelled.
- **Infrastructure interventions:** Flood defences, reservoirs, and channel modifications are not explicitly represented.

8.3 Limited Extreme Tail Data

The reliability of GEV tail estimates is inherently constrained by the number of simulated storms. With a typical catalogue of 200–500 storms, the empirical basis for return periods beyond 50–100 years is weak. The model addresses this partially by flooring exceedance probabilities at 1% (100-year cap), but users should exercise caution when interpreting return levels for very rare events.

8.4 Poisson Independence Assumption

The term structure model assumes that flood events arrive as a homogeneous Poisson process, implying independence between successive events. In reality, flood clustering may occur due to persistent weather patterns (e.g. sequences of Atlantic depressions) or slow-draining catchments where successive storms compound. This could lead to underestimation of multi-year cumulative flood probabilities.

8.5 Multiplicative Noise Model

The noise term $\varepsilon_{g,s} \sim U(0.85, 1.15)$ introduces $\pm 15\%$ variability in gauge response. This is a simplification; empirical studies suggest that gauge response variability may be heteroscedastic (variance increasing with level) and non-symmetric. The uniform distribution also has bounded support, which prevents the generation of truly extreme outlier responses.

8.6 Parameter Defaults

When gauge flood threshold data are missing, the model falls back to default values (Flood Alert = 3.0 m, Flood Warning = 4.0 m, Severe Flood Warning = 5.0 m). These defaults are representative of medium-sized Thames catchment gauges but may be inappropriate for smaller watercourses or upland catchments with different hydrological characteristics. Users should ensure that CDM data are complete before relying on model outputs for pricing or risk decisions.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

“Does the model do what it says on the tin?”

Yes. The model fits a Generalised Extreme Value distribution to simulated peak water levels at river gauge locations and produces (i) annual exceedance probabilities at standard depth thresholds, (ii) return levels for 2–100 year return periods, and (iii) Poisson term structures for PRS pricing. All stated outputs are produced, validated against EA flood frequency estimates, and consumed by downstream models as documented.

9.2 Q2 — Accuracy and Stability

“Does the model provide timely, accurate and stable results?”

The MLE fitting procedure is deterministic for a given input sample; repeated runs with the same storm catalogue produce identical GEV parameters. Bootstrap confidence intervals (1,000 replicates) quantify estimation uncertainty. Anderson–Darling goodness-of-fit tests reject the GEV null at the 5% level for fewer than 5% of Thames catchment gauges. Return levels are benchmarked against EA/N-RFA published flood frequency estimates, with discrepancies exceeding 20% triggering calibration review. The model completes full catchment processing in under 5 seconds.

9.3 Q3 — Resource Usage

“Does the model consume excessive computational resources?”

No. MLE fitting via `scipy.stats.genextreme.fit()` is $O(n)$ per gauge where n is the number of storm realisations (typically 200–500). Term structure computation is $O(1)$ per gauge per severity level. A complete catchment run (25 gauges $\times$ 500 storms) completes in approximately 2–4 seconds on commodity hardware. Memory usage is proportional to the storm catalogue size and does not exceed 50 MB for typical deployments.

9.4 Q4 — Override Controls

“Are there controls over manual overrides?”

The model exposes one configuration override: `force_gumbel`, which constrains the shape parameter $\xi = 0$. This flag is version-controlled and logged at execution time. No manual override of fitted GEV parameters (μ , σ , ξ) or computed exceedance probabilities is supported. Exceedance probabilities are floored at 1% programmatically, preventing user intervention in tail extrapolation. All configuration changes are captured in the model audit trail.

9.5 Q5 — Output Consumption

“How is the output consumed for risk decision-making?”

The model’s outputs serve three downstream decision processes: (a) hazard curve JSON files are consumed by the PRS pricing engine (MKM-PRS-001) for swap spread calculation, (b) return levels feed the property hazard curve model for property-level flood probability assessment, and (c) exceedance probabilities are displayed in gauge-level risk reports used by the Model Risk Committee for portfolio risk oversight. The outputs are not used directly for regulatory capital calculation but inform the risk assessment model that contributes to capital overlays.

9.6 Q6 — Model Chaining

“Is the model’s output used as input to another model?”

Yes. The model occupies a central position in the model chain. Its direct downstream consumers are: (i) PRS Analytical Pricing Engine (MKM-PRS-001), which uses term structures; (ii) Property Hazard Curve module, which uses gauge exceedance probabilities; and (iii) Flood Risk Model (MKM-FLOOD-001), which uses hazard curve points for loss estimation. Upstream, the model consumes gauge responses from the Storm-Gauge Forward Model (MKM-STORM-002) and storm catalogues from the Storm Intensity Distribution (MKM-STORM-001).

9.7 Q7 — Weaknesses

“Does the model have weaknesses and limitations of use?”

Yes; these are documented in detail in Section 8. The principal weaknesses are: (i) the stationarity assumption, which precludes climate change scenarios; (ii) limited reliability of tail estimates beyond the 100-year return period; (iii) the Poisson independence assumption, which ignores flood clustering; and (iv) simplified gauge response model that omits antecedent conditions, spatial rainfall variability, and infrastructure effects.

9.8 Q8 — Monitoring

“What type of monitoring is required?”

The following monitoring regime is recommended: (a) annual recalibration against updated storm catalogues; (b) comparison of fitted GEV parameters against previous year’s values, with a 10% drift threshold triggering review; (c) QQ plot review for any newly instrumented gauges; (d) return period benchmarking against EA estimates following material updates to the NRFA dataset; (e) bootstrap confidence interval width tracking to detect degradation in estimation precision.

9.9 Q9 — Overall Risk

“Is the overall model risk acceptable?”

Conditionally acceptable. As a Tier 1 model with high materiality, the GEV Hazard Curve Model carries significant model risk arising primarily from the stationarity assumption and tail extrapolation uncertainty. These risks are partially mitigated by the conservative probability floor (1% minimum),

the `force_gumbel` option for small samples, and annual backtesting. The unmitigated risk of climate-driven non-stationarity should be addressed in the next model development cycle (non-stationary GEV extension) and is flagged as an open remediation item.

10 Change History

11 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation of GEV Hazard Curve Model	D. Kelly
18-Feb-2026	1.1	Added Section 9: Governance Validation Questions (Chapter 5)	D. Kelly